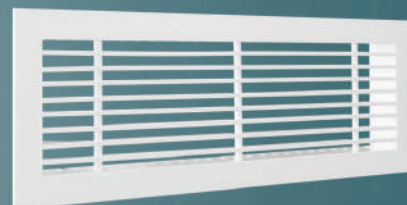




TECNALCO

CHAPTER 3



LINEAR BAR GRILLES

CHAPTER-3

LINEAR BAR GRILLES

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TECNALCO Linear bar grilles and registers provide a high standard of quality, solid construction and quiet, controlled performance.

Features:

- Extruded aluminium anodised linear bars with aerofoil cross section for minimum turbulence.
- For side wall or sill, for supply or return air in heating, cooling or ventilating applications, available in a variety of core-styles, borders and fastening methods.
- They combine architectural beauty with outstanding performance and versatility with fixed bar type blades parallel to the grille's longer dimension.
- For perfect, continuous line usage on a large variety of applications, joining strips are provided in proper lengths with no extra cost.

Types available:

Grille Model	0°		15° - 1 Way		15° - 2 Way	
	6 mm	12.5 mm	6 mm	12.5 mm	6 mm	12.5 mm
RLG	A	A	A	A	N	A
RLG-D	A	A	A	A	N	A
RLG 'TS'	A	A	A	A	N	A
RLG-D-'TS'	A	A	A	A	N	A
SLG	A	A	A	A	N	A
SLG-D	A	A	A	A	N	A
● A → indicates AVAILABLE ● N → indicates NOT AVAILABLE						

- Registers have opposed blade dampers for minimum disturbance of air stream.
- Auxiliary frames available on request for neat and separable mounting in plaster and other wall materials.
- Sponge gasket is provided on request for air tight fixing applications.

Standard sizes available:

Width of the Grille 'W'	Length of the Grille In Number of Pieces 'L'		
	1 pc.	2 pcs.	3 pcs.
50 mm	≤ 4 met.	> 4 met.	> 6 met.
100 mm	≤ 4 met.	> 4 met.	> 6 met.
150 mm	≤ 3.5 met.	> 3.5 met.	> 5 met.
200 mm	≤ 3.5 met.	> 3.5 met.	> 5 met.
250 mm	≤ 3 met.	> 3 met.	> 4.5 met.
300 mm	≤ 2.5 met.	> 2.5 met.	> 4.5 met.
350 mm	≤ 2 met.	> 2 met.	> 4.5 met.

N = Nominal size = Duct size

A = Neck size of the grille

B = Overall size of the grille

where

$N = L \times W$

$A = (L - 10) \times (W - 10) = x \times y$

$B = (x + 60) \times (y + 60) = (L + 50) \times (W + 50)$

Fw = Frame width = 30 mm

Fd = Frame depth = 24 mm / 47 mm

B1p = Linear bar pitch = 6 mm / 12.5 mm

Bp = Blade pitch = 19 mm

Dp = Damper pitch = 25 mm

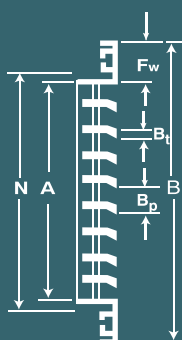
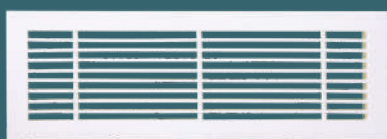
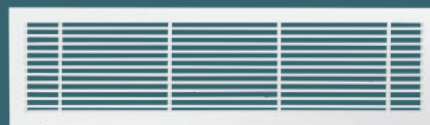
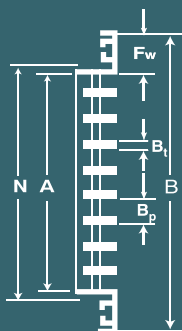
Dd = Damper depth = 36 mm

Frame thickness = 1.2mm

RLG - 0°

TECNALCO model RLG - 0° is a return air grille with fixed profiled linear blades of 0° with 3 mm thickness, set at 12.5 mm or 6 mm pitch.

Fd = Frame depth = 27 mm
Fw = Frame width = 30mm



RLG - 15° - 1 Way

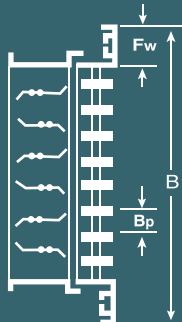
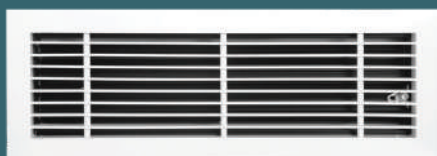
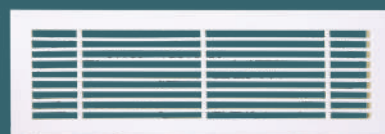
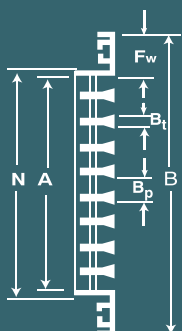
TECNALCO model RLG - 15° - 1 Way is a return air grille with fixed profiled linear blades of 15° - 1 way with 3 mm thickness, set at 12.5 or 6 mm pitch.

Fd = Frame depth = 27 mm
Fw = Frame width = 30mm

RLG - 15° - 2 Way

TECNALCO model RLG - 15° - 2 Way is a return air grille with fixed profiled linear blades of 15° - 2 way set at 12.5 mm pitch.

Fd = Frame depth = 27 mm
Fw = Frame width = 30mm



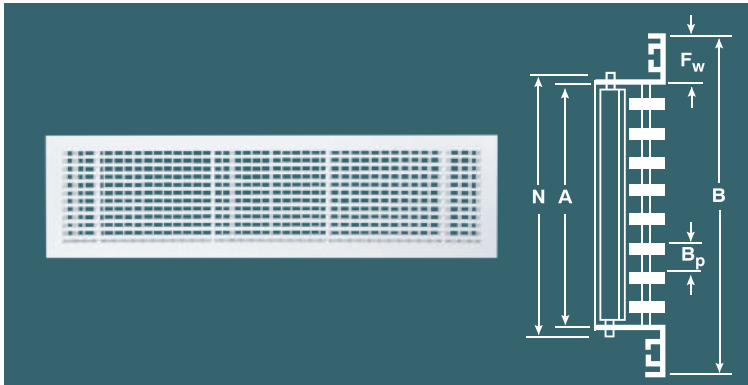
RLG-D - 0°

TECNALCO model RLG-D - 0° is a supply/ return air grille, same as RLG-0° with an opposed blade type volume control damper. The volume control damper can be operated with a key or a screw driver from the face of the grille.

Overall Grille depth = 27 + 36 mm

Note:

- Choice of core styles
- Bar deflection = 0°, 15°- 1 way, 15°- 2 way
- Bar spacing = 6 mm, 12.5 mm
- Optional mounting frames and other accessories are available on request.



SLG - 0°

TECNALCO model SLG - 0° is a supply /return air grille with front fixed profiled 0° linear blades and one set of rear vertical individually adjustable aerofoil blades.

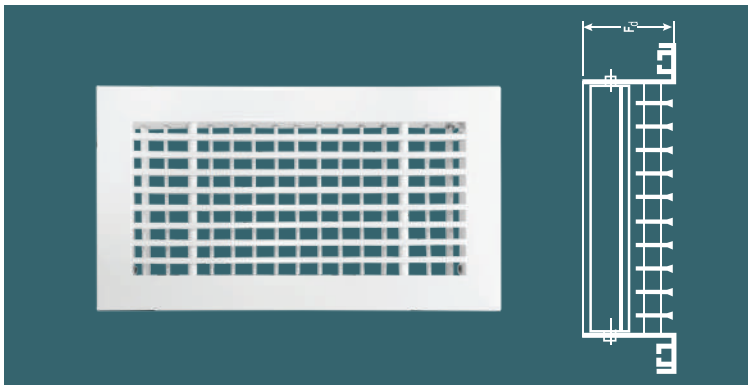
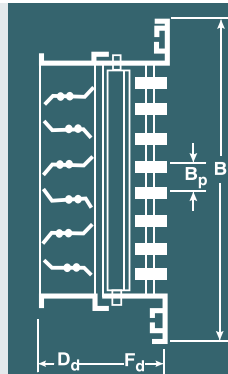
Overall grille depth = 47 mm
Fw = Available size 30mm, 25mm, 20mm, 15mm, 12mm

SLG -D - 0°

TECNALCO model SLG-D -0° is a supply air grille same as SLG -0° with opposed blade type volume control damper.

Opposed blade damper can be operated either with a screw driver or a key from the face of the grille.

Overall grille depth = 47 + 36 mm



SLG - 15° 2 Way

TECNALCO model SLG - 15° 2 way is a supply /return air grille with front fixed profiled 15° linear blades and one set of rear vertical individually adjustable aerofoil blades.

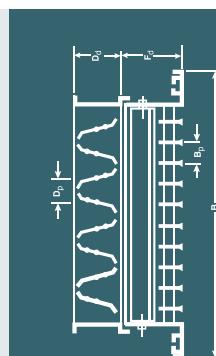
Overall grille depth = 47 mm
Fw = Available size 30mm, 25mm, 20mm, 15mm, 12mm

SLG -D - 15° 2Way

TECNALCO model SLG-D -15° 2way is a supply air grille same as SLG - 15° with opposed blade type volume control damper.

Opposed blade damper can be operated either with a screw driver or a key from the face of the grille.

Overall grille depth = 47 + 36mm



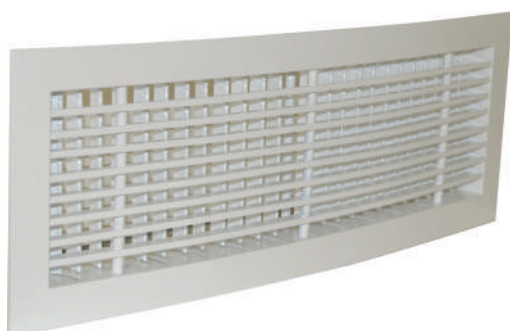
Note:

- All the above combinations are available with '15°- 1 Way' as well.

CURVED LINEAR BAR GRILLES

Curved linear bar grilles are available in 15° 1way and 15° 2way in both SLG and RLG models

EXTERNAL CURVED LINEAR BAR GRILLES

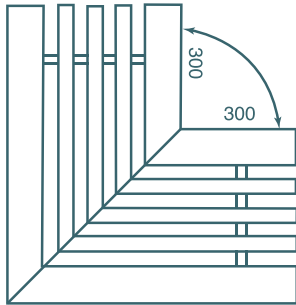


INTERNAL CURVED LINEAR BAR GRILLES

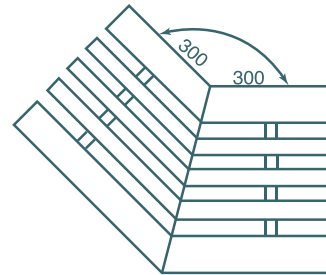


SUPPLY, RETURN, EXTRACT LINEAR BAR GRILLES AND AND REGISTERS

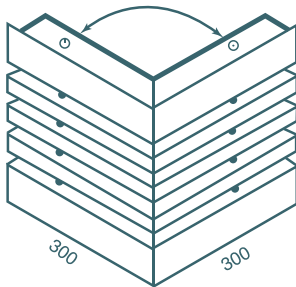
Variety of corners are available in linear bar grilles of models, RLG & SLG types - 0°; 15° -1 way; 15° - 2 Way



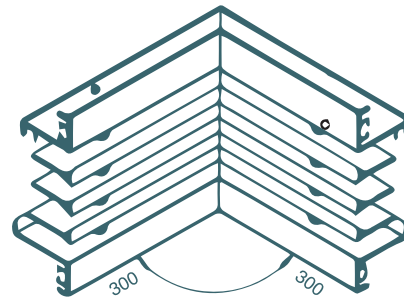
A 90° corner sill or ceiling mounted.



A 135° corner sill or ceiling mounted.



A 90° external corner sill or wall mounted.



A 90° internal corner sill or wall mounted.

NOTE

- Wall or sill mounted type 135° external or internal corners are also available on request.
- For positive alignment of adjacent sections alignment strips provided with each Linear Grille

Linear Bar Grilles of models RLG & SLG types



A → with End Cap (one side DA)



B → with End Flange (one side DB)



C → with Open Ends



E → with both side End Caps/Flanges (EA/EB)



RLG-0° -DA

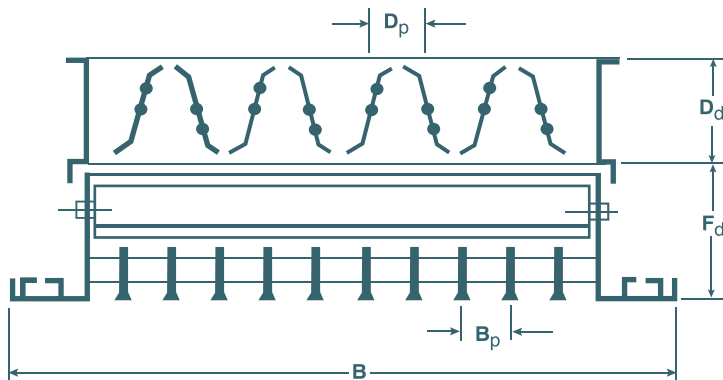
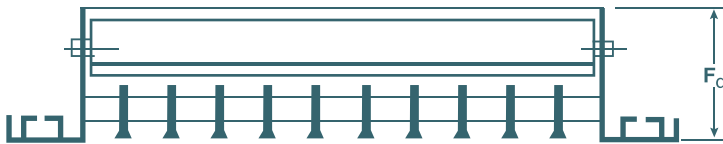
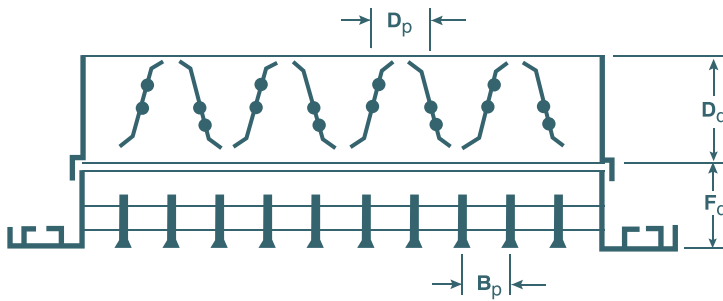
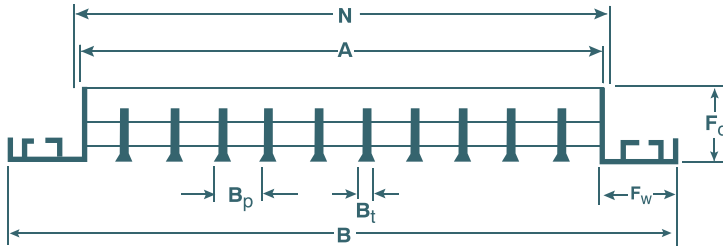


RLG-0° -C



RLG-0° -DA

CONSTRUCTIONAL DETAILS AND DIMENSIONS



Technical Data:

For models RLG types

N = Nominal size = L x W

A = (L - 10) x (W - 10) = x x y

B = (x + 60) x (y + 60) =
(L + 50) x (W + 50)

Fw = Frame width = 30 mm

*Fd = Frame width = 27 mm

Bp = Linear bar pitch =
6 mm / 12.5 mm

Bt = 5 mm 15° - 2 Way & 15° - 1 Way

Bt = 3 mm 0°

Dp = Damper pitch = 25 mm

Dd = Damper depth = 36 mm

Overall grille depth = 27 + 36 mm

*Fd = Frame depth for TS frame
= 47 mm

For models SLG types

N = Nominal size = L x W

A = (L - 10) x (W - 10) = x x y

B = (x + 60) x (y + 60) =
(L + 50) x (W + 50)

Fw = Frame width = 30 mm

Fd = Frame depth = 47 mm

Bp = Linear bar pitch =
6 mm / 12.5 mm

Bpv = Pitch of the vertical blade
= 19 mm

Bt = 5 mm 15° - 2 Way & 15° - 1 Way

Bt = 3 mm 0°

Dp = Damper pitch = 25 mm

Dd = Damper depth = 36 mm

Overall grille depth = 47 + 36 mm

- Operation of the Damper with a key (or) screw driver from the face of the grille, as shown in the figure.

ENGINEERING AND PERFORMANCE DATA

- A wide variety of core styles available.
- The core styles shown here can be used with or without border frames.
- The linear bar spacing (Bar Pitch) at a distance of 6 mm and 12.5 mm are available. (For 15°- 2 way, Bar Pitch 6 mm is not available)
- The supporting bars are placed at a distance of 300 mm (maximum) from each other.

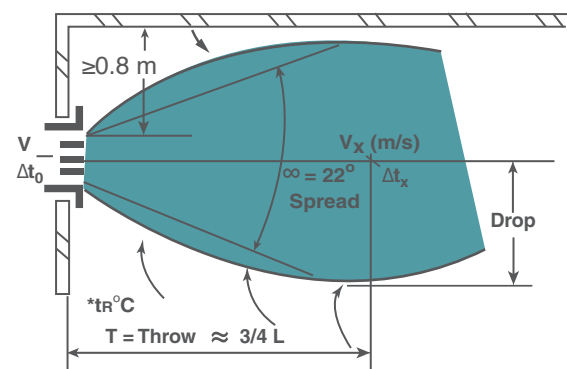
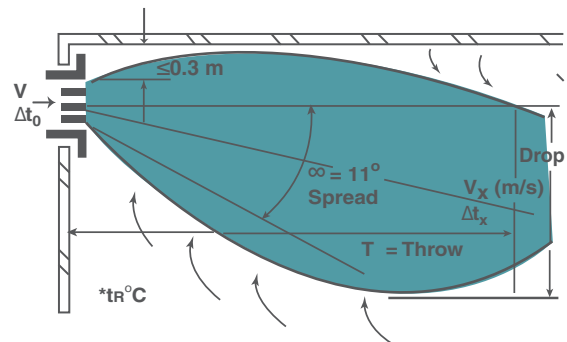
Notes on Selection

- The throw is normally selected up to 3/4 of the room width.
- For a large air volume where the throw is more than 3/4 of the room width distributing the air stream over several outlets will reduce the throw.
- The drop of the selected outlet plus 1.8 met. is the minimum grille or register height from the floor as shown in the figs.
- From the selection diagrams the size of the grille can be selected taking into account the throw, the velocity, pressure loss and noise level of the grille.
- For supply grilles and registers, in addition to air flow, air throw and spread characteristics are the principal factors for selection.
- In order to obtain long throw and narrow air pattern use a deflection between 0° or 15° angle. For shorter throw and wide air pattern use 15° - 2 Way angle Linear Bar grille.
- Performance data shown in the selection charts on the following pages is based on 0°, 15°-1 Way and
- 15°-2 Way Linear Bar Grilles with widths from 50 mm to 300 mm of 1 mt. lengths.

Nomenclature:

A_k	= Effective free area in m ²
V	= Supply air volume m ³ /sec
V_k	= Effective jet velocity m/sec
V_x (max)	= Maximum terminal velocity at distance 'x' . m/sec
V_x aver	= V_x (max) x 0.3 m/sec
X or T	= Throw in metres
Δt_x	= Maximum temperature difference between core and room temperature
$\Delta t_0 = \Delta t_z$	= Temperature difference between supply air temperature and room air temperature
$\frac{\Delta t_x}{\Delta t_z}$	= Temp quotient
$t_R^\circ\text{C}$	= Room temperature ($^\circ\text{C}$)

Type RLG -with ceiling effect



Type RLG-without ceiling effect

V_t	= Terminal velocity in m/sec.
ΔP_t	= Partial pressure in Pa
N.C.	= Noise criteria

ENGINEERING AND PERFORMANCE DATA

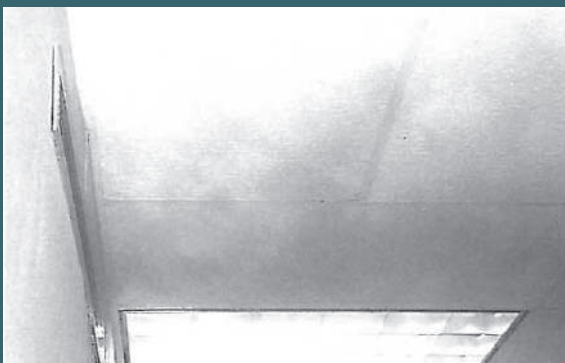
Discharging of air pattern can be observed from the smoke test results.



Model: RLG-D-TS-15°2 Way 1000 x 100 mm
Discharging of air below the ceiling (with ceiling effect)



Model: RLG-D-TS-15°2 Way 1000 x 200 mm
Discharging of air below the ceiling (without ceiling effect)



Model: RLG-D-TS-15°2 Way 1000 x 200 mm
Discharging of air below the ceiling (with ceiling effect)



Model: RLG-D-TS-15°2 Way 1000 x 300 mm
Discharging of air below the ceiling (without ceiling effect)

Smoke tests conducted by ETL Testing Laboratories Inc. An ADC certified lab.

ENGINEERING AND PERFORMANCE DATA

Notes on Selection

- When volume control damper is partially closed for balancing purposes or for final air flow control, in addition to pressure drop and sound corrections, the throw pattern will be reduced between 10 and 18 % depending on the amount of throttling. Pressure drop will increase accordingly.
- The sound level of a supply grille is in direct ratio to the velocity of the air pressure through it.
- Air passing through a properly selected outlet grille will not add any appreciable noise to the sound level of the existing system.
- In order for a new air conditioning system to perform to the designers plans and specifications, it must be properly balanced to deliver the required amount of air through each diffuser.
- Correct ceiling heights must be observed in order to prevent air stream to drop into the occupied zone which is generally about 1.8m. above floor level.

ENGINEERING AND PERFORMANCE DATA

Test Results from ETL

The tests were conducted, by the ETL Testing Laboratories, on the selected sizes of the supply Linear Bar Grilles. The grilles had adjustable opposed blade dampers attached to their inlets. The dampers were set to full open position. The test results include Total pressure, Area factor, Throw pattern and Sound power level.

Test method : The grilles were tested in accordance with Air Diffusion Council Test Code for Grilles, Registers and Diffusers ADC 1062: GRD-84. The grilles were installed in the ADC facility and supplied with measured volumes of air. The static pressure was measured 1-1/2 duct diameters upstream of the grille inlet.

Air volume q_v : Air Volumes are given in Lt/sec.
The throw values given in the tables are related to the Linear Bar Grilles placed below the ceiling. (Without ceiling effect).
Three throw values are given for each q_v to the corresponding terminal velocities V_t .

Throw T : If the Linear Bar Grille is mounted nearer to the ceiling, these throw values should be increased by a factor 1.4
Minimum Throw → 0.75 m/sec. V_t
Middle Throw → 0.50 m/sec. V_t
Maximum Throw → 0.25 m/sec. V_t

NC values: Noise data is given in terms of NC levels. These have been derived from sound power level data (dB re 10-12 watts) by subtracting 10 dB (Room effect) from the sound power levels

Outlet size : Outlet sizes are mentioned on the left side of each row in terms of 1 mt. x grille width within a tolerance of $\pm 10\%$

Linear Bar Grilles Correction factors for other lengths:

L (m)	1	2	3 - 5	6 - 10
Throw (m)	x 1	x 1.05	x 1.1	x 1.15
N.C.	+ 0	+ 1	+ 4	+ 8

ENGINEERING AND PERFORMANCE DATA

Sound Test Data

MANUFACTURER: TECNALCO

PRODUCT: Linear Wall Grille - 1 mt. x 300 mm

N.C. / Inlet Total Pressure 'Pa'

Octave Band Center Frequency Hertz	One Meter x 300 mm				
	Discharge Sound Power Level Lw dB re 10 ⁻¹² Watt				
	1	2	3	4	5
125	43.5	49.0	52.0	55.5	58.0
250	36.5	41.5	43.5	47.0	50.0
500	33.5	38.5	41.0	44.5	47.0
1000	27.0	33.0	36.5	40.5	44.0
2000	19.5*	26.0	29.5	34.0	40.5
4000	21.5*	21.5*	24.0*	28.0	36.0
8000	27.5*	27.5*	27.5*	27.5*	30.5*
Total Air Volume L/s	300	350	400	450	500
Inlet Total Pressure, Pa	13.2	19	26	35	43
**Noise Criteria	17	22.5	25	29	33

* Sound power level data marked with an asterisk has reached ambient levels in the test room. Actual levels are less than or equal to the levels indicated.

** Noise Criteria ratings were determined by subtracting 10 dB (room effect) from the Sound Power Levels.

Area Factor A_k

MANUFACTURER: TECNALCO OUTLET/INLET TYPE & SIZE : 305 mm diameter												
PRODUCT: 1 mt. x 300 mm. Wall Linear Grille								NECK AREA: 729 cm ²				
FLOW METER EQUIPMENT: 305 mm Metering Station 213 mm. Sharp Edged Orifice								ANEMOMETER: Omega HH-30 TYPE & SERIAL NUMBER: 4000-90-011924				
RUN NO.	READINGS							CALCULATIONS				
	Flow Meter P Pa	DISCHARGE VELOCITY IN m/sec.					STATIC PRESSURE Pa	FLOW L/s	NECK VEL. m/s	NECK V.P. Pa	TOTAL PRESS. Pa	AREA FACTOR (A_k)
		1	2	3	4	AVE.						
1	82.14	1.83	1.73	1.57	1.63	1.69	2.99	300	4.11	10.20	13.19	0.178
2	57.25	1.52	1.47	1.32	1.32	1.41	1.99	250	3.43	6.97	8.96	0.177
3	37.34	1.27	1.17	1.07	1.02	1.13	1.24	200	2.74	4.48	5.72	0.177
AVERAGE												0.177

ENGINEERING AND PERFORMANCE DATA

Throw Test Data

Product: Wall Linear Grille 1 Metre x 300 mm
Run No 1 Air Volume 300 L/s

Distance from Outlet (m)													
cm	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0
2.5		1.96	1.50	1.09	0.94	0.94	0.79	0.61	0.61	0.43	0.38	0.30	0.28
7.5		1.32	0.89	0.84	0.74	0.69	0.64	0.51	0.51	0.36	0.33	0.28	0.25
15		1.45	0.69	0.79	0.76	0.71	0.64	0.53	0.51	0.41	0.33	0.28	-
22		1.35	0.64	0.76	0.79	0.71	0.66	0.56	0.51	0.43	0.36	0.30	-
30		1.22	0.56	0.64	0.64	0.56	0.51	0.46	0.36	0.38	0.33	0.30	0.30
45		2.57	1.27	1.07	0.84	0.71	0.64	0.53	0.38	0.46	0.41	0.36	0.30
60		1.04	1.55	1.19	0.97	0.81	0.64	0.48	0.43	0.38	0.36	0.33	0.28
90		-	-	0.28	0.36	0.33	0.30	0.28	0.25	0.25	0.25	-	-
120				-	-	-	-	-	-	-	-		

Velocity in m/s.

Run No. 2 Air Volume 250 L/s

Distance from Outlet (m)													
cm	0	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0	4.5	5.0	5.5	6.0
2.5		1.52	1.32	1.02	0.81	0.74	0.66	0.53	0.48	0.41	0.33	0.30	0.25
7.5		1.04	0.81	0.74	0.66	0.61	0.53	0.48	0.43	0.38	0.30	0.28	0.25
15		1.02	0.66	0.64	0.64	0.58	0.53	0.48	0.43	0.36	0.30	0.28	-
22		0.99	0.66	0.64	0.64	0.61	0.56	0.51	0.46	0.38	0.30	0.25	-
30		0.84	0.48	0.46	0.48	0.41	0.38	0.36	0.33	0.28	-	-	-
45		2.03	1.17	0.89	0.76	0.61	0.48	0.41	0.36	0.30	0.25	0.25	0.25
60		0.94	1.45	1.12	0.79	0.64	0.41	0.33	0.30	0.28	-	-	-
90		-	-	-	0.28	-	-	-	-	-			-
120				-	-	-	-	-	-	-	-		

Velocity in m/s.

ENGINEERING AND PERFORMANCE DATA

Technical data for models RLG & SLG types

Selection Tables: For Area Factor A_k

Effective free area for models RLG & SLG types - 0° & 15° - 1 Way

When opposed blade damper at 100% open position.
For bar pitch $B_p = 6\text{mm} / 12.5\text{mm}$

L x W mm	Bar Pitch	
	6mm $A_k \text{ m}^2$	12.5mm $A_k \text{ m}^2$
1000 x 50	0.02	0.029
1000 x 100	0.041	0.059
1000 x 150	0.063	0.093
1000 x 200	0.09	0.129
1000 x 250	0.115	0.164
1000 x 300	0.144	0.199

NOTE:

- K factor = 0.58 ~ 0.68 for $B_p = 12.5\text{mm}$
- K factor = 0.45 ~ 0.55 for $B_p = 6\text{mm}$

Selection Tables: For Area Factor A_k

Effective free area for models RLG & SLG types - 15° - 2 Way

When opposed blade damper at 100% open position.
For bar pitch $B_p = 12.5\text{mm}$

L x W mm	Bar Pitch	
	6mm	12.5mm $A_k \text{ m}^2$
1000 x 50	-	0.024
1000 x 100	-	0.054
1000 x 150	-	0.085
1000 x 200	-	0.115
1000 x 250	-	0.145
1000 x 300	-	0.177

NOTE:

- K factor = 0.54 ~ 0.6 for $B_p = 12.5\text{mm}$

ENGINEERING AND PERFORMANCE DATA

Selection tables for Linear Bar Grilles
'RLG' and 'SLG' types of models 0° and 15° - 1 Way
Bp = Bar pitch = 6 mm

Nominal width (mm)	Litres / Second / Meter									
	50	100	150	200	250	300	350	400	450	500
50 $A_K = 0.02$	P_t (Pa) Throw (m)	2.2 1.2-3.0-4.3	9.0 2.5-4.3-6.0	19.0 4.3-5.0-7.5	34.0 5.0-6.0-8.5	54.0 5.6-6.8-10.0				
	NC	15	20	23	30	36				
100 $A_K = 0.041$	P_t (Pa) Throw (m)	1.2 1.2-3.0-4.0	5.5 2.5-4.1-5.5	12.0 4.2-5.0-7.2	20.0 5.0-6.0-8.2	32.0 5.6-6.6-9.5	47.0 6.2-7.5-10.6			
	NC	15	18	20	25	34	38			
150 $A_K = 0.063$	P_t (Pa) Throw (m)	1.0 1.0-2.5-3.7	3.0 2.2-4.0-5.3	7.0 4.0-4.7-6.8	13.0 5.0-6.0-8.0	20.0 5.3-6.3-9.0	29.0 6.0-7.5-10.0	38.0 7.5-9.3-11.0	52.0 8.0-9.5-12.5	
	NC	<15	15	20	23	28	30	35	40	
200 $A_K = 0.09$	P_t (Pa) Throw (m)	0.8 0.6-2.1-3.5	2.5 1.8-3.5-5.0	5.0 3.7-4.3-6.7	8.0 4.3-5.7-7.8	13.0 5.0-6.2-8.7	18.0 5.6-6.8-9.7	25.0 6.8-8.7-11.0	30.0 7.5-8.5-11.5	37.0 8.0-9.3-12.5
	NC	<15	15	20	20	20	25	31	35	38
250 $A_K = 0.115$	P_t (Pa) Throw (m)			4.0 3.0-3.7-6.2	7.0 3.7-5.0-7.4	12.0 5.0-5.6-8.5	17.0 4.8-6.2-9.0	23.0 6.5-8.1-10.5	30.0 7.2-8.2-11.0	37.0 8.0-9.0-12.0
	NC			<15	15	20	23	29	32	35
300 $A_K = 0.144$	P_t (Pa) Throw (m)				6.0 1.8-3.7-6.5	10.0 3.7-5.0-7.5	14.0 4.3-5.6-8.5	20.0 6.0-7.5-10.0	27.0 6.8-8.0-10.5	37.0 7.5-8.5-11.0
	NC				<15	15	20	26	28	32
										47.0 8.0-9.5-12.5
										39
										47.0 8.0-9.5-12.5
										37

ENGINEERING AND PERFORMANCE DATA

Selection tables for Linear Bar Grilles
'RLG' and 'SLG' types of model 0° and 15° - 1 Way
Bp = Bar pitch 12.5 mm

Nominal width (mm)		Litres / Second / Meter									
		50	100	150	200	250	300	350	400	450	500
50 $A_k = 0.029$	P_t (Pa)	1.8	7.2	16.0	29.0	45.0					
	Throw (m)	1.0-2.5-3.5	2.0-3.5-5.0	3.5-4.0-6.0	4.0-5.0-7.0	4.5-5.5-8.0					
100 $A_k = 0.059$	NC	<15	<15	18	25	31					
	P_t (Pa)	1.0	4.5	10.0	17.0	27.0	39.0				
	Throw (m)	1.0-2.5-3.3	2.0-3.5-4.5	3.5-4.0-5.8	4.0-5.0-6.8	4.5-5.5-7.8	5.0-6.0-8.5				
150 $A_k = 0.093$	NC	<15	<15	<15	20	29	33				
	P_t (Pa)	0.7	2.5	6.0	11.0	17.0	24.0	32.0	43.0		
	Throw (m)	0.8-2.0-3.0	1.8-3.2-4.5	3.2-3.8-5.5	4.0-4.8-6.5	4.3-5.2-7.5	4.8-6.0-8.0	6.0-7.5-9.0	6.5-7.5-10.0		
200 $A_k = 0.129$	NC	<15	<15	<15	18	23	25	30	35		
	P_t (Pa)	0.5	2.0	4.5	7.0	11.0	15.0	21.0	25.0	31.0	
	Throw (m)	0.5-2.0-3.0	1.5-3.0-4.0	3.0-3.5-5.5	3.5-4.8-6.3	4.0-5.0-7.0	4.5-5.5-7.8	5.5-7.0-9.0	6.0-7.0-9.5	6.5-7.5-10.0	
250 $A_k = 0.164$	NC	<15	<15	<15	<15	15	20	26	30	30	
	P_t (Pa)			3.5	6.0	10.0	14.0	19.0	25.0	31.0	39.0
	Throw (m)			3.5-3.0-5.0	3.0-4.0-6.0	4.0-4.5-6.8	4.0-5.0-7.5	5.5-6.5-8.8	5.8-6.6-9.0	6.5-7.2-9.8	7.0-8.0-10.5
300 $A_k = 0.199$	NC			<15	<15	<15	18	24	27	30	34
	P_t (Pa)				5.0	8.0	11.5	17.0	23.0	31.0	39.0
	Throw (m)				1.5-3.0-5.3	3.0-4.5-7.0	3.5-4.5-7.0	5.0-6.2-8.5	5.5-6.5-8.8	6.0-7.0-9.5	6.5-7.8-10.0
	NC				<15	<15	15	21	23	27	32

ENGINEERING AND PERFORMANCE DATA

Selection tables for Linear Bar Grilles

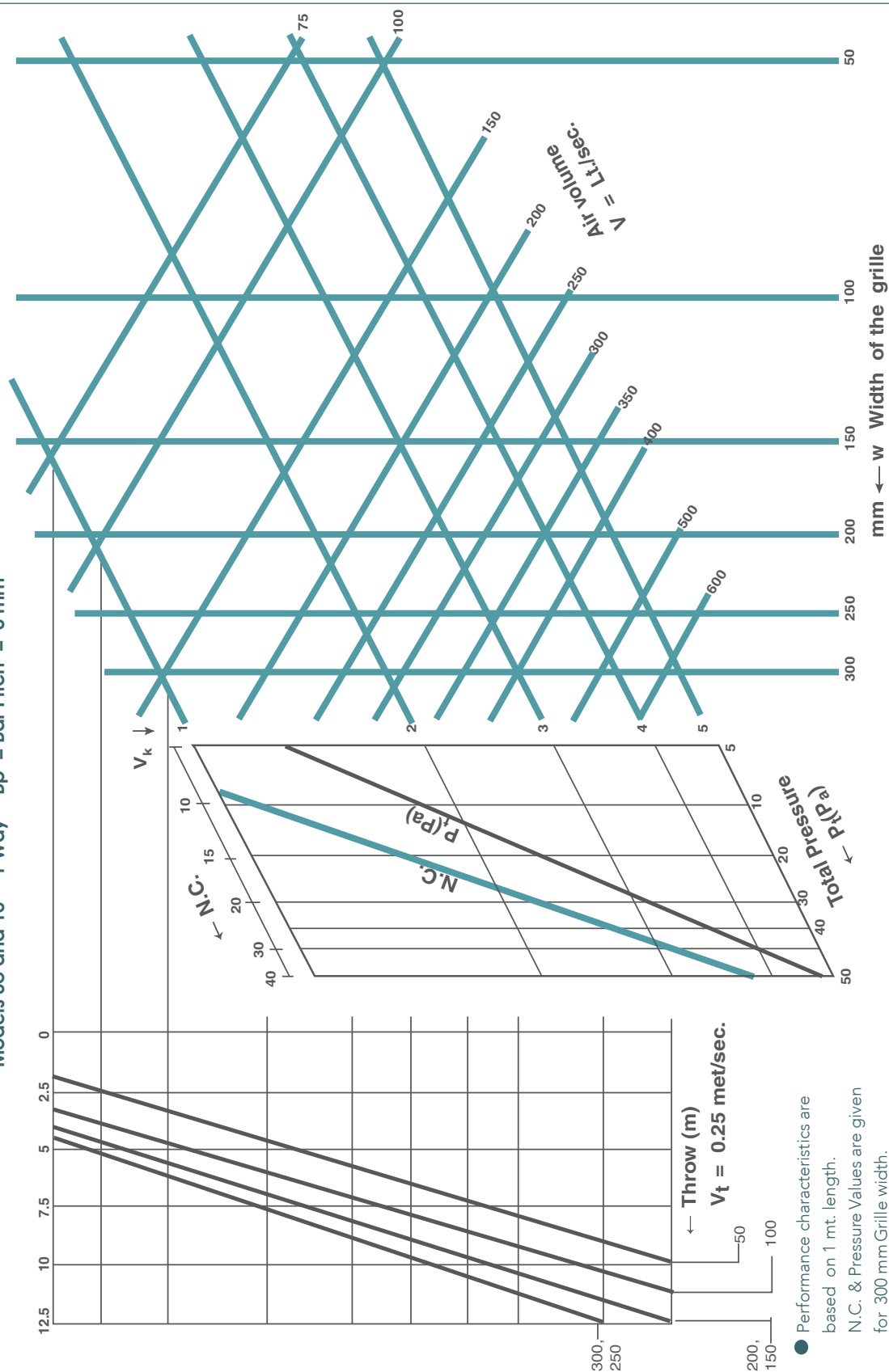
'RLG' and 'SLG' types of model 15° - 2 Way

Bp = 12.5 mm

Nominal width (mm)		Litres / Second / Meter									
		50	100	150	200	250	300	350	400	450	500
50 $A_k = 0.024$	P_t (Pa)	2.0	8.0	18.0	32.0	50.0					
	Throw (m)	1.0-2.5-3.5	2.0-3.5-5.0	3.5-4.0-6.0	4.0-5.0-7.0	4.5-5.5-8.0					
100 $A_k = 0.054$	P_t (Pa)	1.2	5.0	11.0	19.0	30.0	43.0				
	Throw (m)	1.0-2.5-3.5	2.0-3.5-5.0	3.5-4.0-6.0	4.0-5.0-7.0	4.5-5.5-8.0	5.0-6.0-8.5				
150 $A_k = 0.085$	P_t (Pa)	0.8	3.0	7.0	12.0	19.0	27.0	36.0	48.0		
	Throw (m)	1.0-2.5-3.5	2.0-3.5-5.0	3.5-4.0-6.0	4.0-5.0-7.0	4.5-5.5-8.0	5.0-6.0-8.5	6.0-7.5-9.0	6.5-7.5-10.0		
200 $A_k = 0.115$	P_t (Pa)	0.6	2.5	5.0	8.0	12.5	17.0	23.0	28.0	35.0	
	Throw (m)	0.5-2.0-3.0	2.0-3.0-4.5	3.5-4.0-6.0	3.5-5.0-7.0	5.0-5.5-8.0	5.0-6.0-8.5	5.5-7.0-9.0	6.0-7.0-9.5	6.5-7.5-10.5	
250 $A_k = 0.145$	P_t (Pa)			4.0	7.0	11.0	16.0	21.0	28.0	35.0	43.0
	Throw (m)			3.0-4.0-6.0	3.0-4.0-6.0	4.0-4.5-7.0	4.0-5.0-8.0	5.5-7.0-9.0	6.0-7.0-9.5	6.0-7.5-10.0	7.0-8.0-10.5
300 $A_k = 0.177$	P_t (Pa)				5.7	9.0	13.0	19.0	26.0	35.0	43.0
	Throw (m)				1.5-3.0-5.5	2.5-4.0-6.0	3.0-4.5-7.0	5.0-6.5-9.0	5.5-7.0-9.5	6.0-7.5-10.0	6.5-8.0-10.0
	NC				< 15	< 15	17	23	25	29	33

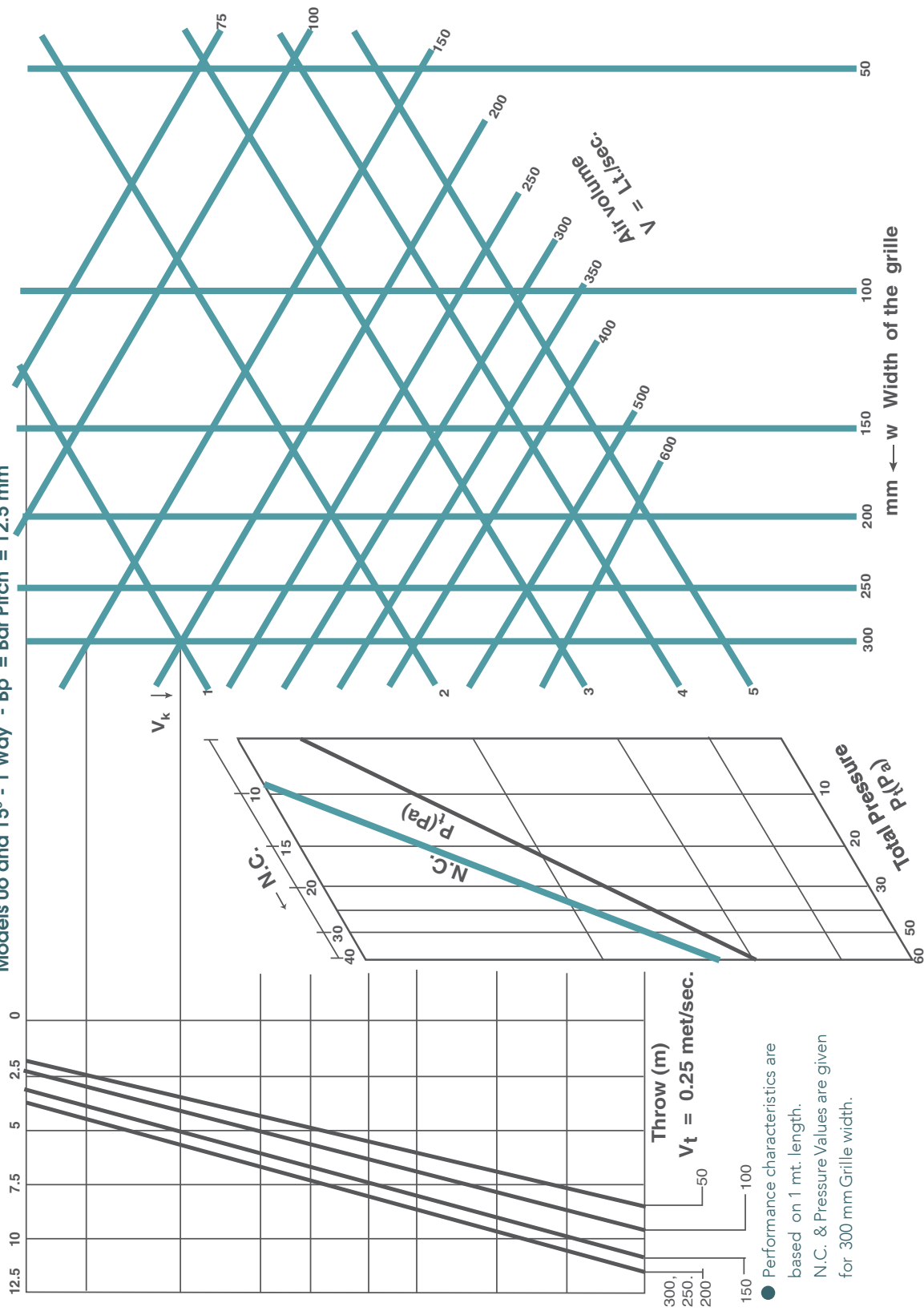
ENGINEERING AND PERFORMANCE DATA

Easy selection graphs for Linear Bar Grilles 'RLG' and 'SLG' types of Models 0o and 15° - 1 Way - Bp = Bar Pitch = 6 mm



ENGINEERING AND PERFORMANCE DATA

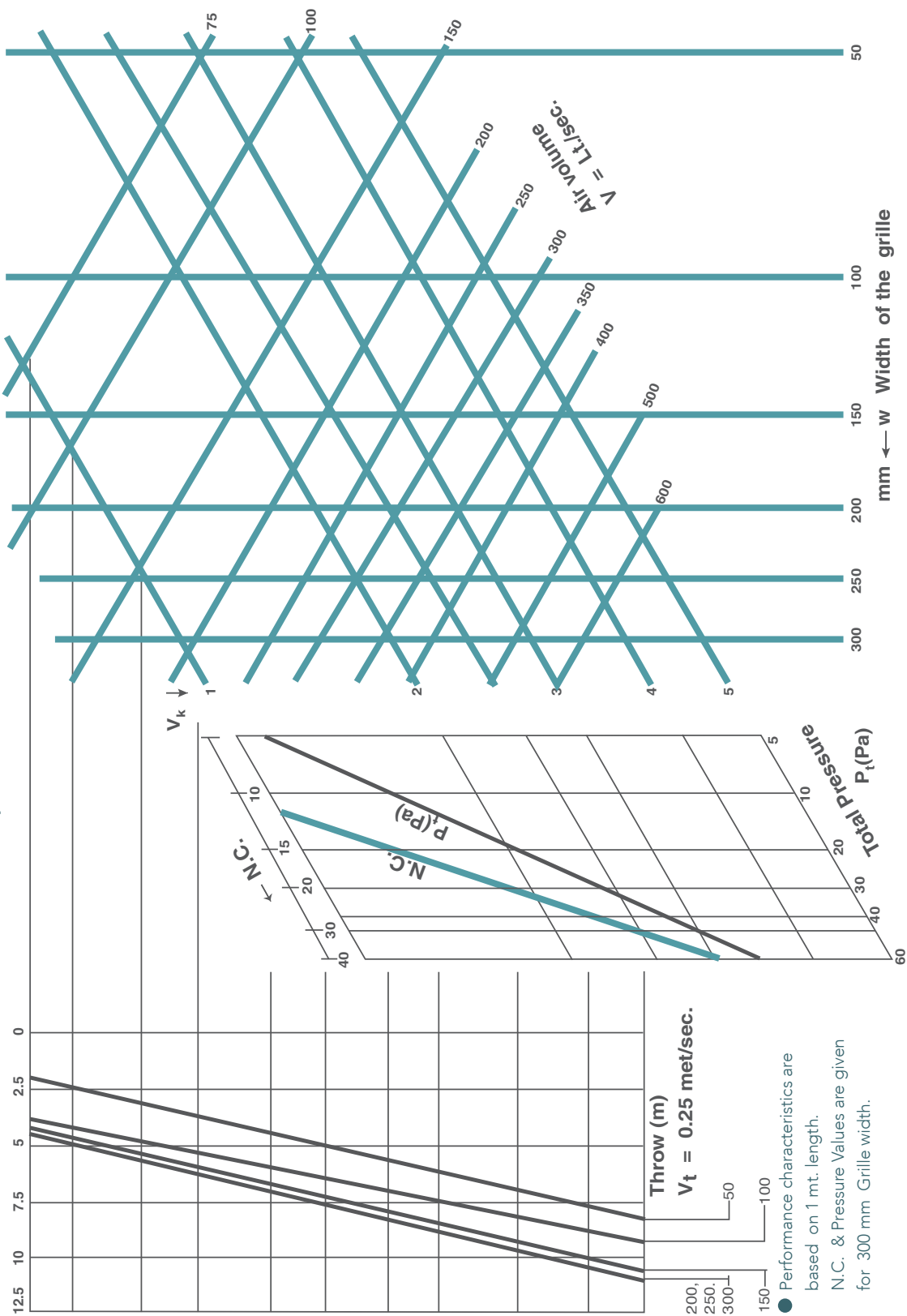
Easy selection graphs for Linear Bar Grilles "RLG" and 'SLG' types of Models 00 and 15° - 1 Way - Bp = Bar Pitch = 12.5 mm



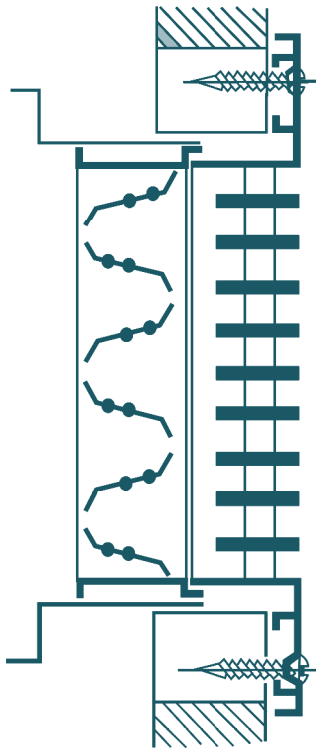
ENGINEERING AND PERFORMANCE DATA

Easy selection graphs for Linear Bar Grilles - RLG and SLG types of Models 15° - 2 Way -

Bp = Bar Pitch = 12.5 mm



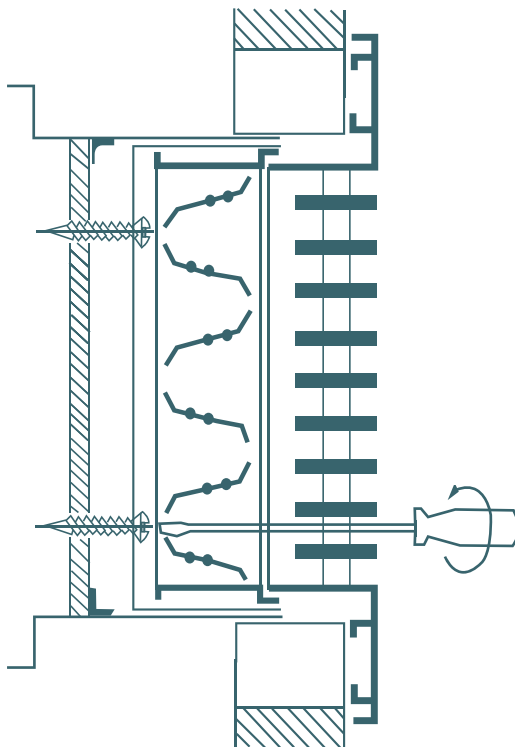
LINEAR BAR GRILLES FIXING DETAILS



Type 'S' fixing:

A counter- sunk screw type fixing

- Recommended for sidewall, ceiling and sill applications.



Type 'C' fixing:

Concealed type fixing with wooden bar

- Recommended for sidewall, and sill applications.

LINEAR BAR GRILLES FIXING DETAILS



Type 'C' Fixing:

Concealed bracket type fixing

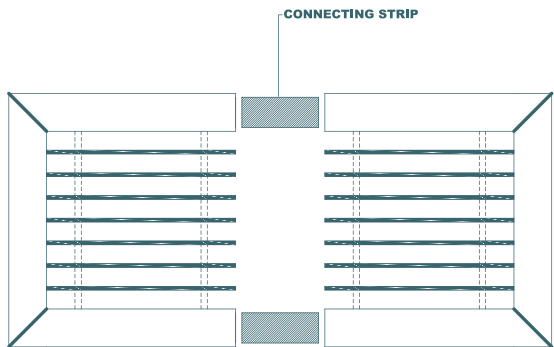
- Recommended for sidewall, ceiling and sill applications

NOTE:

Standard supply is **Type 'C' fixing**.

- Alignment or joining clips are provided for all standard type sections of TECNALCO model Linear Bar Grilles, when supplied with open ends for continuous installation.

Two Long pices linear bar grill joining diagram



Available finishes:

- Natural anodised aluminium
- Powder coated colour finish. (Mention RAL colour number)
- Mill finish aluminium

Available finishes for the accessories :

- Mill finish aluminium
- Powder coated in black maat finish

Ordering Data:

Specify

- | | | |
|----------------------------|-------------------|-------------------|
| 1. TECNALCO type and model | 3. Quantity | 5. Surface finish |
| 2. Nominal size - L x W | 4. Type of fixing | 6. Remarks |

Example:

1.	2.	3.	4.	5.	6.
RLG-D-0° 'TS' frame	3000x 150 mm	12 nos.	Type 'C' fixing	Powder coated RAL 9016	D — Damper Black matt



TECNALCO

Quality Air Distribution Since 1981

ALUMINIUM TECHNICAL ENGINEERING FACTORY

Manufacturer of high quality Air distribution devices and Duct accessories

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